Arkaraj Mukherjee:
Homework 7

```r
library("tidyverse")
getwd()
```

```
## [1] "/home/anon1/work/uni-stuff/sem_II/intro_to_statistics_and_computation_with_da
```

1. The likelihood function here is,

$$\mathcal{L}(X_1, \ldots, X_k; p_1, \ldots, p_k) = \binom{n}{X_1, \ldots, X_k} p_1^{X_1} \cdots p_k^{X_k}$$

with parameter space $\{(p_1, \ldots, p_k) \in \mathbb{R}^k : \forall i, p_i \geq 0 \text{ and } \sum_{i=1}^k p_i = 1\}$. The log likelihood l in this case will be

$$l(X_1, \ldots, X_k; p_1, \ldots, p_k) = \log(n!) - \sum_{i=1}^k \log(X_i!) + \sum_{i=1}^k X_i \log(p_i)$$

It suffices to maximize $\sum_{i=1}^k X_i \log(p_i)$ instead. It will suffice to show that this evaluated at $p_i = X_i/n$ is an upper bound, so we do that:

$$\sum_{i=1}^k X_i \log(p_i) - \sum_{i=1}^k X_i \log(X_i/n) = \sum_{i=1}^k X_i \log(np_i/X_i)$$

$$\leq \sum_{i=1}^k X_i(np_i/X_i - 1) = \sum_{i=1}^k (np_i - X_i) = n\sum_{i=1}^k p_i - \sum_{i=1}^k X_i = n - n = 0$$

Here we have used the inequality $\log(t) \leq t - 1$ which is valid for all positive t, this stems from the fact that $e^{t-1} \geq (t-1) + 1 = t$. We have thus shown that,

$$\sum_{i=1}^k X_i \log(p_i) \leq \sum_{i=1}^k X_i \log(X_i/n)$$

so we are done.

2. (a) To show that $\mathbb{E}[S^2] = \sigma^2$, we use the standard unbiased estimator for sample variance, $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$. First, we expand the sum of squares:

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) = \sum_{i=1}^n X_i^2 - 2\bar{X}\sum_{i=1}^n X_i + n\bar{X}^2$$

Since $\sum_{i=1}^{n} X_i = n\bar{X}$, this simplifies to:

$$\sum_{i=1}^{n} X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \sum_{i=1}^{n} X_i^2 - n\bar{X}^2$$

Now we take the expectation. Recall that $\mathbb{E}[X^2] = \mathrm{Var}(X) + (\mathbb{E}[X])^2$. Thus, $\mathbb{E}[X_i^2] = \sigma^2 + \mu^2$. For the sample mean, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, so $\mathbb{E}[\bar{X}^2] = \frac{\sigma^2}{n} + \mu^2$.

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = \sum_{i=1}^{n}\mathbb{E}[X_i^2] - n\mathbb{E}[\bar{X}^2]$$

$$= n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2 = (n-1)\sigma^2$$

Finally, multiplying by $\frac{1}{n-1}$ yields:

$$\mathbb{E}[S^2] = \frac{1}{n-1}\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = \frac{1}{n-1}(n-1)\sigma^2 = \sigma^2$$

(b) Let Z be a random variable with finite mean and variance. By definition, the variance of Z is $\mathrm{Var}(Z) = \mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right] = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2$. Since the expected value of a non-negative random variable (a squared real value) must be non-negative, we have $\mathrm{Var}(Z) \geq 0$. Therefore, $\mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 \geq 0 \implies \mathbb{E}[Z^2] \geq (\mathbb{E}[Z])^2$.

Equality fails when $\mathrm{Var}(Z) > 0$, meaning Z is not a constant almost surely. For example, let Z be the outcome of a fair 6-sided die. $\mathbb{E}[Z] = 3.5$, so $(\mathbb{E}[Z])^2 = 12.25$. However, $\mathbb{E}[Z^2] = (1^2 + 2^2 + \cdots + 6^2)/6 \approx 15.17$. Clearly, $15.17 > 12.25$.

We apply this inequality by setting $Z = S$, the sample standard deviation. This gives $\mathbb{E}[S^2] \geq (\mathbb{E}[S])^2$. From part (a), we established that $\mathbb{E}[S^2] = \sigma^2$. Substituting this in yields $\sigma^2 \geq (\mathbb{E}[S])^2$. Since the standard deviation S is strictly non-negative by definition ($S = \sqrt{S^2}$), its expectation $\mathbb{E}[S]$ must also be non-negative. Taking the square root of both sides gives $\sigma \geq \mathbb{E}[S]$.

Equality fails when $\mathrm{Var}(S) > 0$. If we draw random samples from a Normal distribution, S^2 follows a scaled Chi-square distribution, meaning S will vary from sample to sample. Hence $\mathrm{Var}(S) > 0$ and $\mathbb{E}[S] < \sigma$, proving S is a biased estimator.

(c) To compute $\mathrm{Var}(S^2)$, we can assume without loss of generality that $\mu = 0$ (since variance is shift-invariant, substituting $X_i' = X_i - \mu$ doesn't change S^2). Thus, $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = \sigma^2$, and $\mathbb{E}[X_i^4] = \gamma$. By definition, $\mathrm{Var}(S^2) = \mathbb{E}[S^4] - (\mathbb{E}[S^2])^2 = \mathbb{E}[S^4] - \sigma^4$.

We expand $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - n\bar{X}^2\right)$. Squaring this gives:

$$(n-1)^2 S^4 = \left(\sum_{i=1}^{n} X_i^2\right)^2 - 2n\bar{X}^2\sum_{i=1}^{n} X_i^2 + n^2\bar{X}^4$$

Now we take the expectation of each term. Because the X_i are i.i.d. with mean 0, cross-terms with odd powers vanish.
First, $\mathbb{E}\left[\left(\sum_{i=1}^n X_i^2\right)^2\right] = \mathbb{E}\left[\sum_{i=1}^n X_i^4 + \sum_{i\neq j} X_i^2 X_j^2\right] = n\gamma + n(n-1)\sigma^4$.
Second, for $2n\bar{X}^2 \sum_{i=1}^n X_i^2$, we write $\bar{X}^2 = \frac{1}{n^2}\left(\sum_{j=1}^n X_j\right)^2$. The expectation simplifies to:

$$\mathbb{E}\left[2n \cdot \frac{1}{n^2}\left(\sum_{j=1}^n X_j\right)^2 \sum_{i=1}^n X_i^2\right] = \frac{2}{n}\mathbb{E}\left[\left(\sum_{j=1}^n X_j^2 + \sum_{j\neq k} X_j X_k\right)\sum_{i=1}^n X_i^2\right]$$

$$= \frac{2}{n}\mathbb{E}\left[\left(\sum_{i=1}^n X_i^2\right)^2\right] = \frac{2}{n}(n\gamma + n(n-1)\sigma^4) = 2\gamma + 2(n-1)\sigma^4$$

Third, for $n^2\bar{X}^4 = n^2 \cdot \frac{1}{n^4}\left(\sum_{i=1}^n X_i\right)^4 = \frac{1}{n^2}\left(\sum_{i=1}^n X_i\right)^4$. Expanding the fourth power and taking expectations (only X_i^4 and $X_i^2 X_j^2$ terms survive):

$$\frac{1}{n^2}\mathbb{E}\left[\sum_{i=1}^n X_i^4 + 3\sum_{i\neq j} X_i^2 X_j^2\right] = \frac{1}{n^2}(n\gamma + 3n(n-1)\sigma^4) = \frac{\gamma}{n} + \frac{3(n-1)}{n}\sigma^4$$

Combining these expected values:

$$(n-1)^2\mathbb{E}[S^4] = (n\gamma + n(n-1)\sigma^4) - (2\gamma + 2(n-1)\sigma^4) + \left(\frac{\gamma}{n} + \frac{3(n-1)}{n}\sigma^4\right)$$

Grouping terms with γ and σ^4:

$$(n-1)^2\mathbb{E}[S^4] = \gamma\left(n - 2 + \frac{1}{n}\right) + \sigma^4\left(n^2 - n - 2n + 2 + 3 - \frac{3}{n}\right)$$

$$= \gamma\frac{(n-1)^2}{n} + \sigma^4\frac{(n-1)(n^2-2n+3)}{n}$$

Dividing by $(n-1)^2$, we get:

$$\mathbb{E}[S^4] = \frac{\gamma}{n} + \frac{n^2-2n+3}{n(n-1)}\sigma^4$$

Finally, subtracting σ^4 gives the variance:

$$\mathrm{Var}(S^2) = \mathbb{E}[S^4] - \sigma^4 = \frac{\gamma}{n} + \left(\frac{n^2-2n+3}{n^2-n} - 1\right)\sigma^4 = \frac{\gamma}{n} - \frac{n-3}{n(n-1)}\sigma^4$$

Taking the limit as $n \to \infty$:

$$\lim_{n\to\infty}\mathrm{Var}(S^2) = \lim_{n\to\infty}\left(\frac{\gamma}{n} - \frac{n-3}{n(n-1)}\sigma^4\right) = 0 - 0 = 0$$

This demonstrates that $\mathrm{Var}(S^2) \to 0$ as $n \to \infty$.

3. To find the exact distribution of $Z = \frac{X_1}{\sqrt{X_2/n}}$, we can use the known formula for the pdf of the ratio of two continuous random variables. Let $Y = \sqrt{X_2/n}$. First, we find the pdf of Y. Since $X_2 \sim \chi^2_n$, its pdf is:

$$f_{X_2}(v) = \frac{1}{2^{n/2}\Gamma(n/2)} v^{n/2-1} e^{-v/2}, \quad v > 0$$

The transformation is $y = \sqrt{v/n}$, which implies $v = ny^2$ and the derivative $dv/dy = 2ny$. Thus,

$$f_Y(y) = f_{X_2}(ny^2) \left| \frac{dv}{dy} \right| = \frac{1}{2^{n/2}\Gamma(n/2)} (ny^2)^{n/2-1} e^{-ny^2/2} (2ny)$$

$$f_Y(y) = \frac{2(n/2)^{n/2}}{\Gamma(n/2)} y^{n-1} e^{-ny^2/2}, \quad y > 0$$

Now, we have $Z = X_1/Y$. Because X_1 and X_2 are independent, X_1 and Y are independent. For $Y > 0$, the standard formula for the pdf of a ratio is:

$$f_Z(z) = \int_0^\infty y f_{X_1}(zy) f_Y(y) dy$$

Substitute $f_{X_1}(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and our derived $f_Y(y)$ into the integral:

$$f_Z(z) = \int_0^\infty y \left(\frac{1}{\sqrt{2\pi}} e^{-(zy)^2/2} \right) \left(\frac{2(n/2)^{n/2}}{\Gamma(n/2)} y^{n-1} e^{-ny^2/2} \right) dy$$

$$f_Z(z) = \frac{2(n/2)^{n/2}}{\sqrt{2\pi}\Gamma(n/2)} \int_0^\infty y^n e^{-\frac{1}{2}y^2(z^2+n)} dy$$

To evaluate this integral, we can luckily use the substitution $u = \frac{1}{2}y^2(z^2+n)$. This means $du = y(z^2+n)dy$, so $ydy = \frac{du}{z^2+n}$. We also have $y^2 = \frac{2u}{z^2+n}$, which implies $y^{n-1} = \left(\frac{2u}{z^2+n}\right)^{(n-1)/2}$.

$$f_Z(z) = \frac{2(n/2)^{n/2}}{\sqrt{2\pi}\Gamma(n/2)} \int_0^\infty \left(\frac{2u}{z^2+n} \right)^{(n-1)/2} e^{-u} \frac{du}{z^2+n}$$

$$f_Z(z) = \frac{2(n/2)^{n/2}}{\sqrt{2\pi}\Gamma(n/2)} \frac{2^{(n-1)/2}}{(z^2+n)^{(n+1)/2}} \int_0^\infty u^{(n-1)/2} e^{-u} du$$

The remaining integral is exactly the definition of the Gamma function, evaluated at $\frac{n+1}{2}$. We substitute $\int_0^\infty u^{(n-1)/2} e^{-u} du = \Gamma\left(\frac{n+1}{2}\right)$ and simplify the constants:

$$f_Z(z) = \frac{2 \cdot n^{n/2} 2^{-n/2} \cdot 2^{(n-1)/2}}{\sqrt{2}\sqrt{\pi}\Gamma(n/2)} \frac{\Gamma\left(\frac{n+1}{2}\right)}{(z^2+n)^{(n+1)/2}}$$

$$f_Z(z) = \frac{\sqrt{2}n^{n/2}}{\sqrt{2}\sqrt{\pi}\Gamma(n/2)} \frac{\Gamma\left(\frac{n+1}{2}\right)}{n^{(n+1)/2}(1+z^2/n)^{(n+1)/2}}$$

$$f_Z(z) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{z^2}{n}\right)^{-\frac{n+1}{2}}$$

This is exactly the pdf of the t-distribution with n degrees of freedom, proving that $Z \sim t_n$.

4. (a)

```
n = 100
x_bar = 125
s = 10
alpha = 0.01
t_star = qt(1 - alpha/2, df = n - 1)
margin_error = t_star * (s / sqrt(n))
ci_99 = c(x_bar - margin_error, x_bar + margin_error)
ci_99

## [1] 122.3736 127.6264

        diff(ci_99)

## [1] 5.252811
```

The 99% confidence interval is roughly $[122.37, 127.63]$.

(b)

```
alpha = 0.05
t_star = qt(1 - alpha/2, df = n - 1)
margin_error = t_star * (s / sqrt(n))
ci_95 = c(x_bar - margin_error, x_bar + margin_error)
ci_95

## [1] 123.0158 126.9842

        diff(ci_95)

## [1] 3.968434
```

The 95% confidence interval is roughly $[123.02, 126.98]$.

Comparing the intervals from part (a) and (b), the 99% confidence interval is strictly wider than the 95% confidence interval. This occurs because achieving a higher level of confidence requires a larger critical t-value, which correspondingly increases the margin of error when the sample size and sample standard deviation are held constant.

5. (a)

```r
data = read_csv("wristneck.csv")
## Rows:  51 Columns:  2
## - Column specification --------------------------------------
## Delimiter:  ","
## dbl (2):  Wrist, Neck
##
## i Use `spec()` to retrieve the full column specification for this
data.
## i Specify the column types or set `show_col_types = FALSE` to
quiet this message.

cor_coeff = cor(data$Wrist, data$Neck)
cor_coeff

## [1] 0.6501159
```

The computed correlation coefficient between the wrist and neck measurements is shown in the R output above.

(b)

```r
model = lm(Wrist ~ Neck, data = data)
coef(model)

## (Intercept)        Neck
##   2.3595004   0.2911116

        1/coef(model)[2]

##     Neck
## 3.435108
```

Using ordinary least squares regression, the best fit estimates for the coefficients are β_0 (the intercept) and β_1 (the slope for the neck measurement), which can be read directly from the output and we see that the waist is almost 3.5 times in circumference as the wrist for our class.